

# MOHAMED A. MOHAMED

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## SUMMARY

Infrastructure engineer specializing in secure cloud platforms, Kubernetes, Infrastructure as Code, CI/CD automation, and AI infrastructure. Designs production-style systems with security, reliability, observability, and operational efficiency built in from the start. Hands-on experience delivering AWS and Kubernetes platforms, custom operators, GitOps workflows, policy enforcement, and scalable infrastructure for containerized and AI workloads.

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## SKILLS

Cloud & Infrastructure: AWS, EKS, EC2, VPC, IAM, S3, ECR, RDS, CloudTrail, GuardDuty, Security Hub, Terraform, CloudFormation

Containers & Platforms: Kubernetes, Docker, Helm, Kustomize, Argo CD, Crossplane, Backstage, Cilium, Istio

DevSecOps & Security: GitHub Actions, OIDC, Trivy, Checkov, Gitleaks, Kyverno, OPA Gatekeeper, NetworkPolicy, RBAC, secrets management

AI Infrastructure: RAG, vLLM, Qdrant, pgvector, FastAPI, LLM inference, GPU infrastructure, prompt-injection controls, AI governance

Observability & Automation: Prometheus, Grafana, Python, Bash, Boto3, Linux, incident response, automated remediation

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## EXPERIENCE

### IT SYSTEMS SPECIALIST | Galoof & Associates | BOSTON, MA | January 2021 - Present

- Supported business-critical technology operations for 9 users, 11 workstations, 3 servers, printers, scanners, networking, backups, email, and tax software.
- Maintained reliable office operations during high-volume tax seasons by resolving daily endpoint, access, application, printing, and connectivity issues.
- Automated recurring maintenance, monitoring, configuration, backup, and compliance tasks to reduce manual intervention and improve consistency.
- Supported identity and productivity platform transitions involving Microsoft 365, Outlook, Entra ID, Intune, and cloud-managed services.
- Created operational checklists and recovery procedures that improved system readiness and reduced disruption during peak business periods.

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## PROJECTS

### LLM RAG Infrastructure Platform | [github.com/SecureCloudOps/llm-rag-infra-platform](https://github.com/SecureCloudOps/llm-rag-infra-platform)

- Built a production-style RAG platform supporting document ingestion, vector retrieval, source-grounded responses, and pluggable LLM providers.
- Deployed containerized AI services to Amazon EKS using GitOps, Kustomize, managed node groups, EBS storage, and isolated workload namespaces.
- Integrated CI security scanning with Trivy, Checkov, and Gitleaks while growing automated test coverage to 25 passing tests.
- Designed the platform to support scalable model inference, persistent vector storage, operational health checks, and repeatable cloud deployment.

### Secure AI Platform Engineering | [github.com/SecureCloudOps/secure-k8s-devsecops](https://github.com/SecureCloudOps/secure-k8s-devsecops)

- Designed an internal developer platform combining Kubernetes infrastructure, service catalog capabilities, secure CI/CD, observability, and an AI-powered engineering assistant.
- Built reusable Terraform infrastructure for networking, EKS, IAM, and ECR with GitHub Actions OIDC, eliminating long-lived cloud credentials from CI/CD.
- Implemented RAG-based infrastructure assistance with source attribution, audit metadata, secret redaction, safe-file filtering, and prompt-injection defenses.
- Established Kubernetes security controls including namespace isolation, least-privilege RBAC, restricted workloads, and deny-by-default network policies.

### KubeBackup Operator | [github.com/SecureCloudOps/database-backup-operator](https://github.com/SecureCloudOps/database-backup-operator)

- Engineered a Kubernetes operator that automates checksum-verified PostgreSQL backups and restores through custom resources.
- Enforced restricted security contexts, least-privilege RBAC, scoped network access, resource limits, and disabled service-account token mounting.
- Added backup health and freshness metrics, AWS S3 support, failure-state reporting, and checksum validation before restoration.
- Achieved 21 passing controller tests and 3 end-to-end Kind tests; published a signed and verifiable container release.

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## CERTIFICATIONS

**AWS Certified SysOps Administrator – Associate | 2025**

**Certified Kubernetes Administrator | 2025**

**CompTIA Security+ | 2025**

**AWS Certified Solutions Architect – Associate | 2024**

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